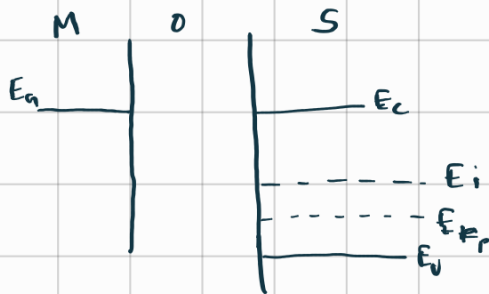


Lab-1)

Problem - 2.1



NMOS:

$$\phi_{nc} = \phi_n - \phi_c \quad \text{where } \phi_c = \phi_{FP}$$

$$\phi_{FP} = \frac{KT}{q} \ln\left(\frac{N_A}{n_i}\right) = 0.0259 \ln\left(\frac{3 \times 10^{20}}{1.45 \times 10^{10}}\right)$$

$\uparrow$
 $T = 300K$
 $= 0.44V$

$\phi_n = -0.55V$. Since, if poly Si is highly doped, the gate level approaches E_c .

$$\phi_{nc} = -0.55 - 0.44 = -0.99V$$

$$V_{FB} = 0.99 - \frac{9.6 \times 10^{-8}}{1.6 \times 10^6} = -1.05V$$

Now, NMOS has p-type substrate. So, the depletion region contains '-ve' charge.

$$Q_{D0} = \left[2 q N_A \epsilon_{Si} |2\phi_{FP}| \right]^{1/2}$$

$$= -3 \times 10^{-7} C/cm^2$$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

$$= \frac{3.9 \times 8.854 \times 10^{-14}}{22 \times 10^{-10}}$$

$$= 1.6 \times 10^{-6} F/cm^2$$

$$Q_{ox} = q N_{ss}$$

$$= 1.6 \times 10^{-19} \times 6 \times 10^{11}$$

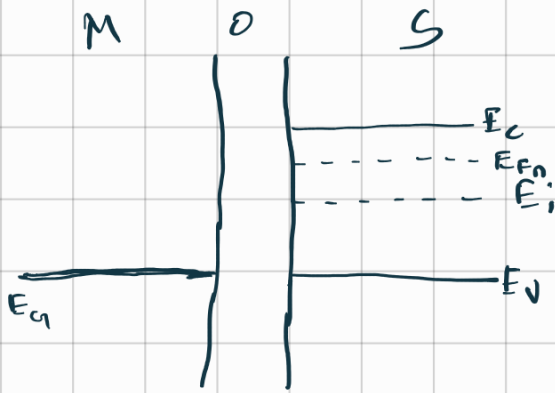
$$= 9.6 \times 10^{-8} C/cm^2$$

$$\epsilon_{Si} = 11.7 \times 8.854 \times 10^{-14}$$

$$V_{T0} = -1.05 + 2 \times 0.44 - \frac{-3 \times 10^{-7}}{1.6 \times 10^{-6}}$$

$$= 0.02 \text{ V}$$

For PMOS:



$$\phi_{Fn} = \frac{kT}{q} \ln\left(\frac{n_i}{N_D}\right)$$

$$= -0.44 \text{ V.}$$

$$\phi_n = 0.55 \text{ V. [since, gate is p+ poly-Si]}$$

$$\phi_{sc} = 0.55 - (-0.44) \text{ V} = 0.99 \text{ V.}$$

$$V_{FB} = 0.99 - 0.06 \text{ V} = 0.93 \text{ V.}$$

$$Q_{Bo} = +\left(2qN_D\epsilon_{Si}2\phi_{Fn}\right)^{1/2}$$

$$= 3 \times 10^{-7} \text{ C/cm}^2$$

$$V_{T0} = 0.93 + 2(-0.44) - \frac{3 \times 10^{-7}}{1.6 \times 10^{-8}}$$

$$= -0.14 \text{ V.}$$

(b) If the n+ poly-Si gate is changed to p+ and p+ poly-Si gate is changed to n+, sign of ϕ_n would flip for each case.

For NMOS:-

$$\phi_n = 0.55 \text{ V}$$

$$\text{and } \phi_{ni} = 0.55 - 0.44 \text{ V} = 0.11 \text{ V}$$

For PMOS:

$$\phi_n = -0.55 \text{ V}$$

$$\text{and } \phi_{ni} = -0.55 - (-0.44)$$

$$= -0.11 \text{ V.}$$

NMOS:

$$V_{FB_{new}} = 0.11 - 0.06$$

$$= \boxed{0.05 \text{ V}}$$

and

$$V_{T_{new}} = 0.05 + 0.88 + 0.19$$

$$= \boxed{1.22 \text{ V}}$$

PMOS:

$$V_{FB_{new}} = -0.11 - 0.06$$

$$= \boxed{-0.17 \text{ V}}$$

$$V_{T_{new}} = -0.17 - 0.88 - 0.19$$

$$= \boxed{-1.24 \text{ V}}$$

Using Molybdenum (M_o) gate ($\Phi_m = 0$) only changes the Φ_{ac} and changes both V_{FB} and V_{T_0} for both NMOS and PMOS.

For NMOS:-

$$\Phi_{ac} = 0 - 0.44 \text{ V}$$

$$= -0.44 \text{ V}$$

$$V_{FB} = -0.44 - \frac{9.6 \times 10^{-8}}{1.6 \times 10^{-6}}$$

$$= \boxed{-0.5 \text{ V}}$$

and

$$V_{T_0} = -0.5 + 2 \times 0.44 - \frac{3 \times 10^{-7}}{1.6 \times 10^{-6}}$$

$$= \boxed{0.57 \text{ V}}$$

For PMOS:-

$$\Phi_{ac} = 0 - (-0.44 \text{ V})$$

$$= 0.44 \text{ V}$$

$$V_{FB} = 0.44 - \frac{9.6 \times 10^{-8}}{1.6 \times 10^{-6}}$$

$$= \boxed{0.38 \text{ V}}$$

and

$$V_{T_0} = 0.38 - 2 \times 0.44 - \frac{3 \times 10^{-7}}{1.6 \times 10^{-6}}$$

$$= \boxed{-0.69 \text{ V}}$$

With M_o gate, V_{T_0} is closer to the desired $\pm 0.4 \text{ V}$ and requires reduced threshold implant. However, M_o is very difficult to use in processing.

d) (a) For NMOS:

$$\Delta V_T = 0.4 - 0.02 \text{ V}$$

$$= 0.38 \text{ V}$$

Therefore, implant of p-type dopant required for positive adjustment

For PMOS:

$$\Delta V_T = -0.4 - (-0.14) \text{ V}$$

$$= -0.26 \text{ V}$$

Need to implant n-type & donors to adjust the threshold voltage

$$N_x = \frac{C_{ox} \Delta V_{T_0}}{q}$$

$$= \frac{1.6 \times 10^{-6} \times 0.38}{1.6 \times 10^{-19}}$$

$$= 3.83 \times 10^{12} \text{ ions/cm}^2$$

$$N_A = \frac{N_x}{x_j} = \frac{3.83 \times 10^{12} \text{ ions/cm}^2}{2.5 \times 10^{-6} \text{ cm}}$$

$$= 1.535 \times 10^{18} \text{ atoms/cm}^3$$

$$N_x = \frac{C_{ox} |\Delta V_{T_0}|}{1.6 \times 10^{-19}}$$

$$= \frac{1.6 \times 10^{-6} \times 0.26}{1.6 \times 10^{-19}}$$

$$= 2.64 \times 10^{12} \text{ ions/cm}^2$$

$$N_D = \frac{N_x}{x_j} = \frac{2.64 \times 10^{12} \text{ ions/cm}^2}{2.5 \times 10^{-6} \text{ cm}}$$

$$= 1.05 \times 10^{18} \text{ atoms/cm}^3$$

(b) For NMOS:

Using the same process as (a)

$$\Delta V_T = 0.4 - 1.22$$

$$= -0.82$$

Needs donor implant.

$$N_x = 8.2 \times 10^{12} \text{ ions/cm}^2$$

and

$$N_D = 3.28 \times 10^{18} \text{ atoms/cm}^3$$

For PMOS:-

$$\Delta V_T = -0.4 - (-1.24)$$

$$= 0.84$$

positive adjustment $\rightarrow$ need acceptor implantation.

$$N_x = 8.4 \times 10^{12} \text{ ions/cm}^2$$

$$\text{and } N_A = 3.36 \times 10^{18} \text{ atoms/cm}^3$$

(v) Using p^+ dope Poly-Si in NMOS and n^+ Poly-Si in PMOS would cause the flatband voltage to increase in both cases. This causes the threshold voltage to go up significantly. Ideally, the desired threshold voltage is $\pm 0.4V$. Using same type of doping in both bulk and gate causes the V_{T_0} in part (ii) be 1.22 (NMOS) and -1.24 (PMOS) which is beyond the supply voltage. Using poly-Si gate with same type of doping as the source and drain tends to result in a closer to desired V_T . Also, Poly-Si gate is used to block the source and drain region implants. So, having the same type doping eases the fabrication process.

Problem-2.3

(i) For $0.13\mu\text{m}$ technology, $V_{dd} = 1.2\text{V}$
 $L = 100\text{nm} = 1 \times 10^{-5}\text{cm}$

NMOS:

$$\text{Horizontal E-field, } E_y = \frac{V_{dd}}{L} = \frac{1.2\text{V}}{1 \times 10^{-5}\text{cm}} = 1.2 \times 10^5 \text{ V/cm.}$$

Since, the horizontal field $E_y > E_{cn} (6 \times 10^4 \text{ V/cm})$

carriers reach saturation velocity in the channel.

So, the NMOS is a short-channel device.

PMOS:- Horizontal E-field $E_y < E_{cp} (24 \times 10^4 \text{ V/cm})$.

Since the holes do not reach saturation velocity in the channel, the PMOS is a long-channel device.

(ii) NMOS:-

For $V_{gs} > V_T (\approx 0.4)$

$$I_{DS} = I_S \exp\left(\frac{q(V_{gs} - V_T - V_{offset})}{nKT}\right) \left(1 - e^{-\frac{qV_{DS}}{KT}}\right) \approx 0.$$

For $V_{gs} \geq V_T$:

$$\text{if } V_{DS} < \frac{(V_{gs} - V_T) \times E_c L}{(V_{gs} - V_T) + E_c L} \quad (\text{linear region})$$

$$\text{we get, } I_{DS} = \frac{W}{L} \cdot \frac{\mu_e \cdot C_{ox}}{\left(1 + \frac{V_{DS}}{E_c L}\right)} \left(V_{gs} - V_T - \frac{V_{DS}}{2}\right) \cdot V_{DS}.$$

$$\text{Here, } W = 100 \times 10^{-7} \text{ cm}$$

$$L = 100 \times 10^{-7} \text{ cm.}$$

$$\mu_e = \frac{\mu_0}{1 + \left(\frac{V_{gs} - V_T}{\theta \cdot t_{ox}}\right)^n} \quad [HSS - 2.21] \text{ effect of vertical field on mobility}$$

$$\theta = 3.6 \times 10^6 \text{ V/cm}$$

$$\eta = 1.85$$

$$t_{ox} = 2.2 \times 10^{-7} \text{ cm}$$

$$\mu_0 = 540 \text{ cm}^2/\text{Vsec}$$

0.13 μm technology

Using the equation we get,

For $V_{gs} = 1.2 \text{ V}$:

$$\mu_e = 267 \text{ cm}^2/\text{Vsec}$$

and $V_{gs} = 0.8 \text{ V}$:

$$\mu_e = 421 \text{ cm}^2/\text{Vsec}$$

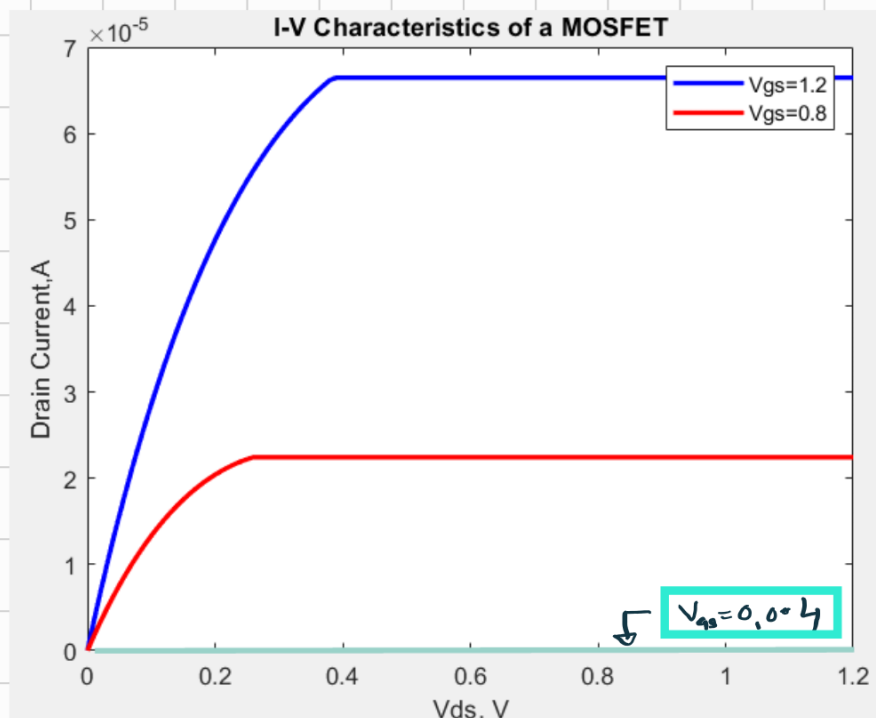
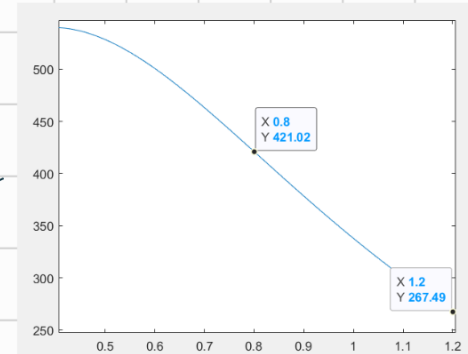
$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = 1.6 \times 10^{-6} \text{ F/cm}^2$$

$$E_c = \frac{2V_{sat}}{\mu_e}; \text{ where } V_{sat} = 1 \times 10^7 \text{ cm}$$

Again,

if $V_{ds} \geq \frac{(V_{gs} - V_T) \times E_c L}{(V_{gs} - V_T) + E_c}$:

$$I_{Ds} = \frac{W}{L} V_{sat} \cdot C_{ox} \frac{(V_{gs} - V_T)^2}{(V_{gs} - V_T) + E_c L}$$



PMOS:- long channel device;

For $V_{GS} < V_T$,

$$I_{DS} = 0$$

For $V_{GS} \geq V_T$

if $V_{DS} < V_{GS} - V_T \rightarrow$ linear region

$$I_{DS} = \frac{k}{2} [2(V_{GS} - V_T) \cdot V_{DS} - V_{DS}^2]$$

$$k = \mu_p \cdot C_{ox}$$

For $V_{GS} = 1.2V$;
 $\mu_p = 70 \text{ cm}^2/\text{Vsec}$. about $\frac{\mu_e}{4}$.

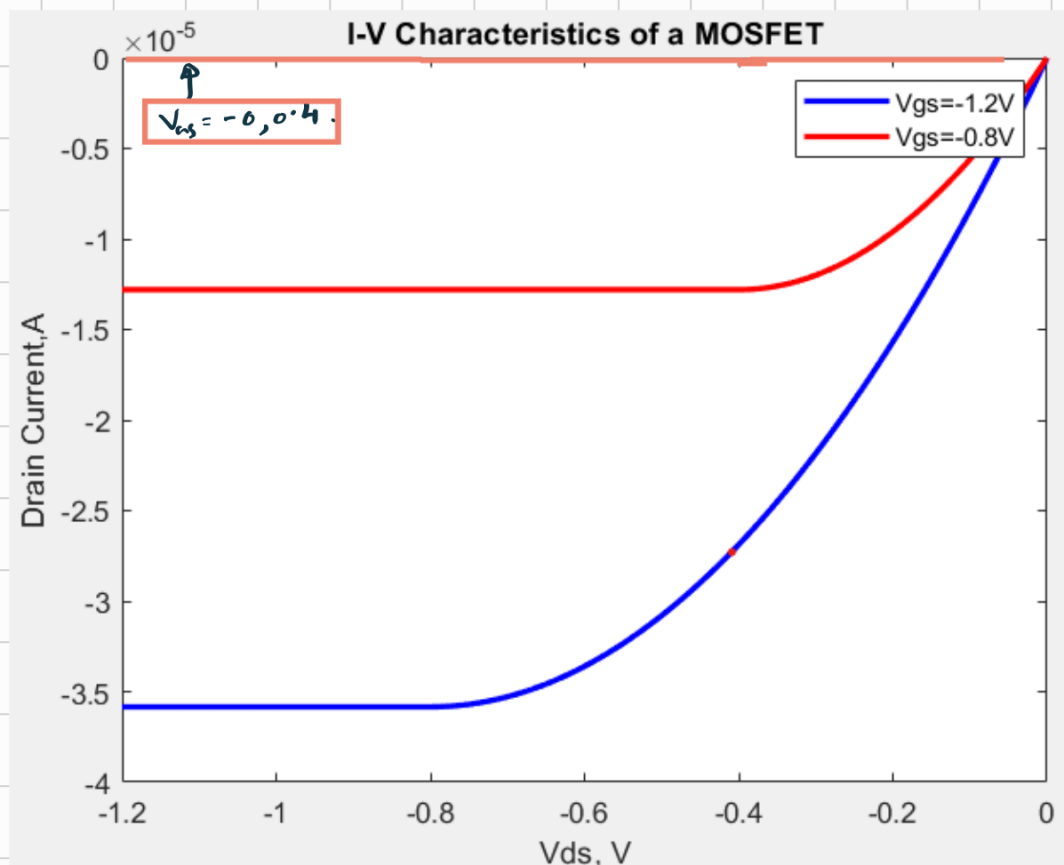
similarly for $V_{GS} = 0.8V$

$$\mu_p \approx 100 \text{ cm}^2/\text{Vsec}$$

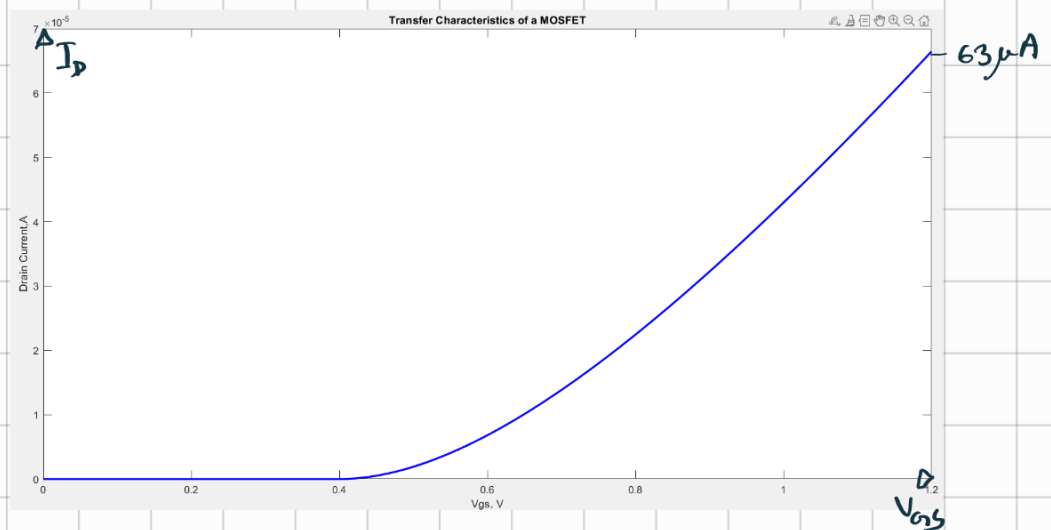
$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = 1.6 \times 10^{-6} \text{ F/cm}^2.$$

If $V_{DS} \geq V_{GS} - V_T$

$$I_{DS} = \frac{k}{2} (V_{GS} - V_T)^2.$$



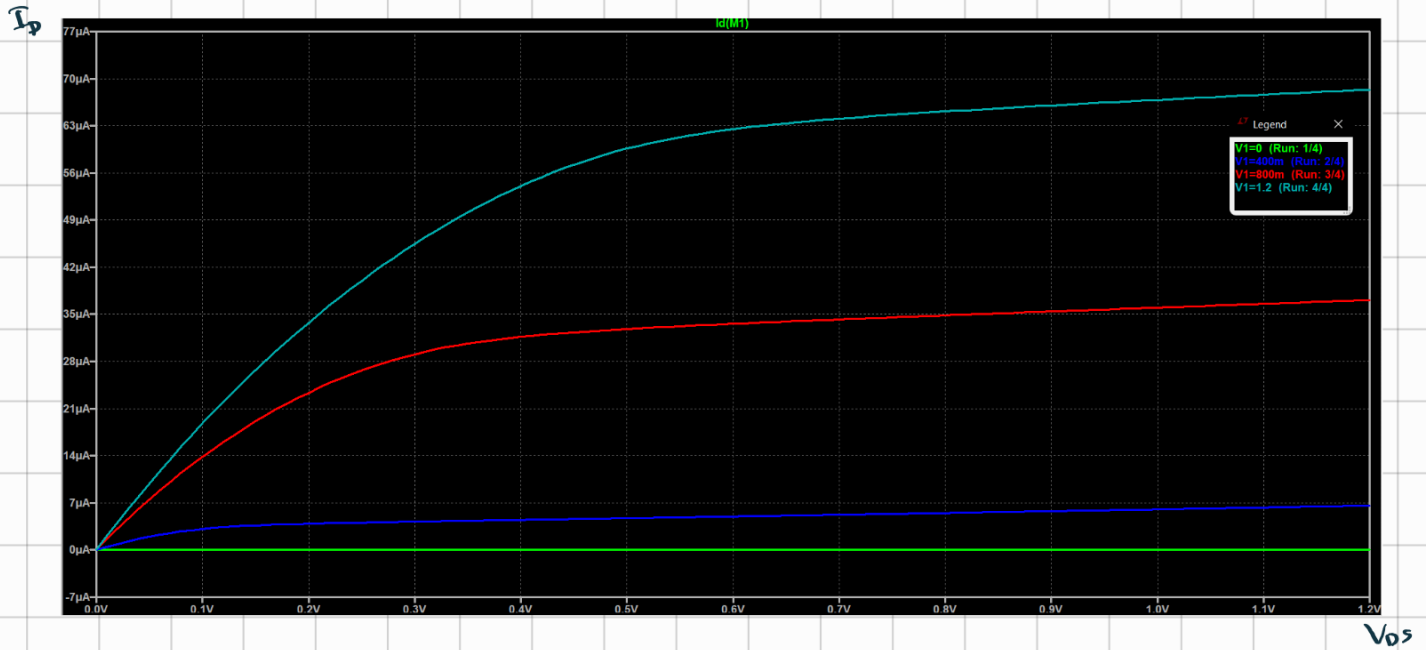
(iii) Using the calculation and equation in part (ii)



The relationship seems to be more linear compared to quadratic in the figure. So, the quadratic model used for long-channel device is not sufficient for the short-channel device due to high field and carrier velocity saturation.

(u) For NMOS:-

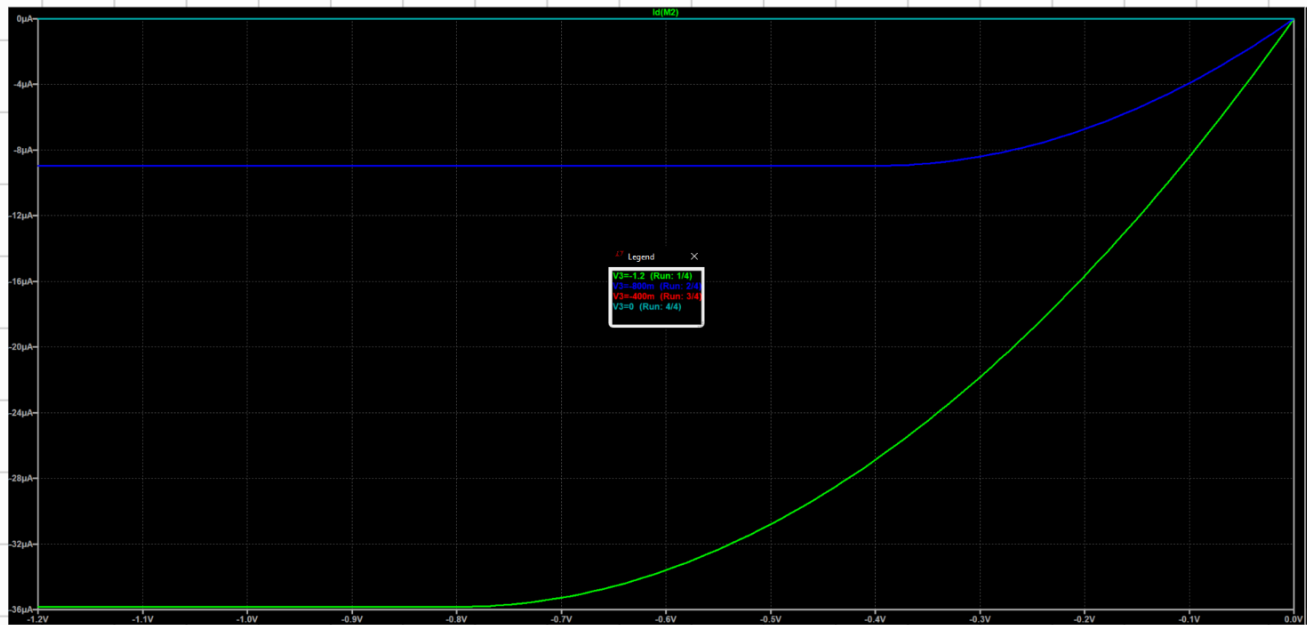
I_D vs V_{DS} for $V_{GS} = 0, 0.4, 0.8$ and 1.2



The simulated model in LTSPICE provides a slightly smaller current compared to the hand-calculations. This is due to the fact that the level 49 SPICE model uses a lot more parameters than our hand-calculation equation to resemble the performance of an actual device.

For PMOS:-

Using $K_p = 112 \mu$ and $V_{t_0} = -0.4V$



The simulation results for the long-channel PMOS device agrees with our hand calculation in part (ii)

For the transfer characteristics of the NMOS:



Again, the simulated result gives us a slightly lower value than the expected hand-calculated value. as expected.

Problem - 2.5

a) The NMOS is on when $V_{GS} > V_T$:

Here, the source is being pulled up.

At $t=0$: $V_x = 0 = V_s$.

$$V_{GS} = 1.2 ; V_T = 0.4 \quad \text{and} \quad V_{DS} = 1 > V_{GS} - V_T \\ (= 1.2 - 0.4 = 0.8)$$

So, the MOSFET will be in saturation. As the source is

pulled-up NMOS goes to cut-off as soon as $V_s = V_x$ reaches the point where, $V_{GS} = V_T$

$$V_G - V_x = V_T$$

$$\Rightarrow V_x = V_G - V_T \\ = 1.2 - 0.4 \\ = 0.8V.$$

So, V_x never reaches $V_D = 1V$ due to this limitation.

(b) Here, $V_D = 0.6V$ & $V_{DS} < V_{GS} - V_T (0.8V) \rightarrow$ linear operation

Here, V_s never reaches the $0.8V$ limit as the source cannot pull-up to a higher voltage than what is supplied ($V_D = 0.6$). So, the current flow stops at $V_x = 0.6V$ as V_{DS} reaches zero.

The LTSPICE simulation shows exactly same results as expected.

